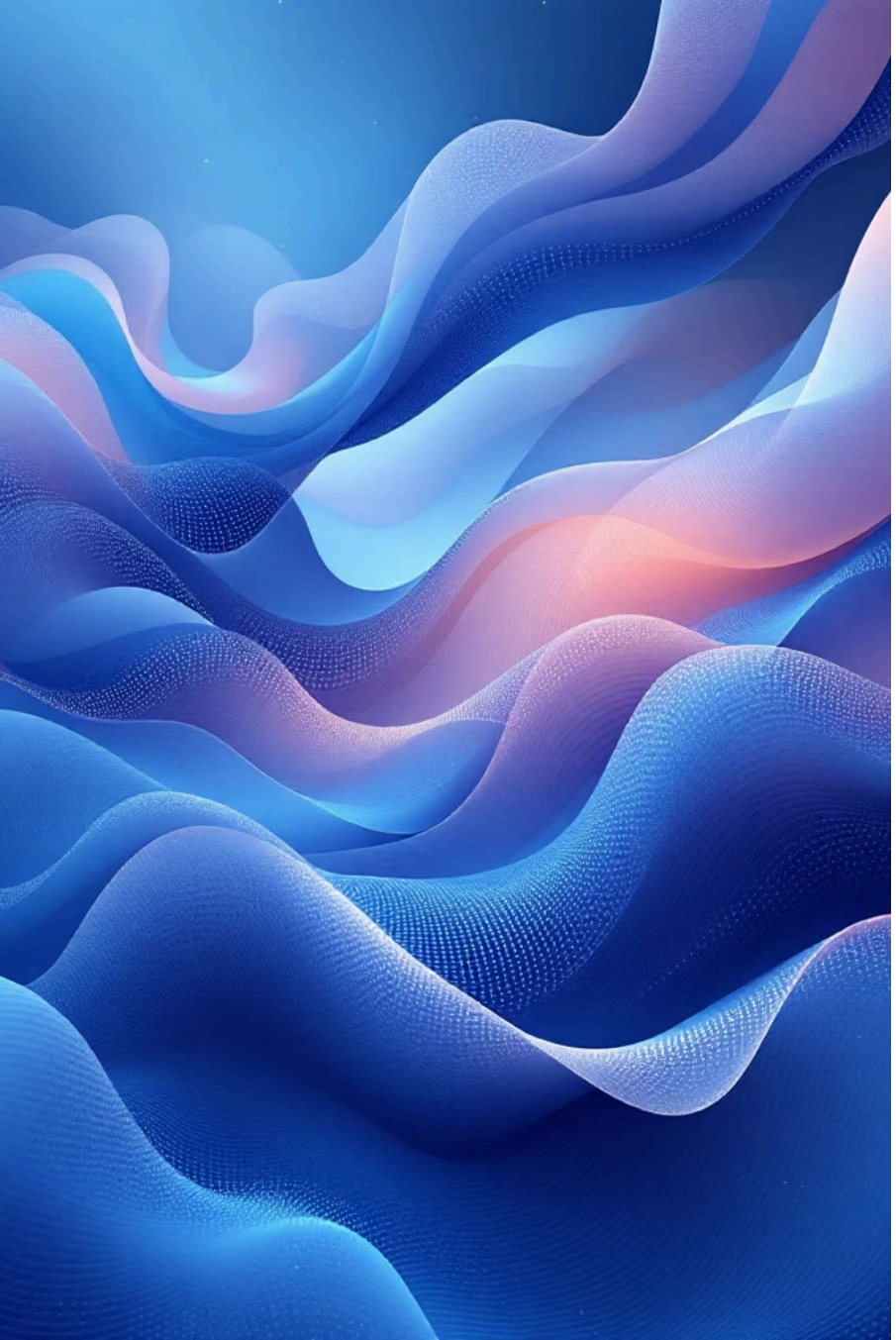

Global Unemployment Analysis

Gender and Age Group Disparities (2014–2024)

PIPELINE: PYTHON · SQL · POWER BI

DECISION MAKERS & RECRUITERS

DATA ANALYST PORTFOLIO

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Context & Business Question

Why this analysis?

Unemployment reveals deep structural inequalities by gender, age, and geography. Decision-makers need a comparative, segmented, and time-based view to guide employment policy.

Core business question

> "To what extent have unemployment inequalities between genders and age groups evolved across countries between 2014 and 2024, and which national profiles concentrate the most critical vulnerabilities?"

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Dataset & Analytical Pipeline

Source: International Labour Organization (ILO)

1134

Observations

189

Countries Covered

11

Years (2014–2024)

Variables: country, gender, age group, annual rates

Analytical pipeline in 3 steps

Python (Jupyter)

Quality audit, cleaning, feature engineering

SQL (MySQL Workbench)

10 autonomous analytical queries

Power BI

3-page decision dashboard, 17 DAX measures

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Quality Audit & Cleaning (Python)

What the audit revealed

Missing values

30 missing values across 2 countries in conflict zones: Palestinian Territories (2023–2024) and Ukraine (2022–2024). Correction method: **Forward Fill (ffill)** — propagates the last known value forward.

Outliers (IQR)

Kept — verifiable economic realities (rates >60% in certain countries). These extremes reflect real crises, not data entry errors.

Column removed

1 column removed: indicator_name — constant value across all 1,134 rows, with no analytical value.

Final Result

0
Missing values

22
Final columns

Clean, consistent dataset ready for SQL analysis and Power BI visualization.

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Feature Engineering — 7 Variables Created

Business variables built to directly answer decision-making questions.

rate_avg_2014_2024

Structural level over 10 years

rate_latest

Current 2024 status

covid_impact

Pandemic shock (2020 vs 2019)

decadal_trend

Raw change over 10 years

trend_direction

Improving / Stable / Worsening

risk_level

Low / Medium / High / Critical (ILO thresholds)

gender_gap

Female - Male gap in 2024

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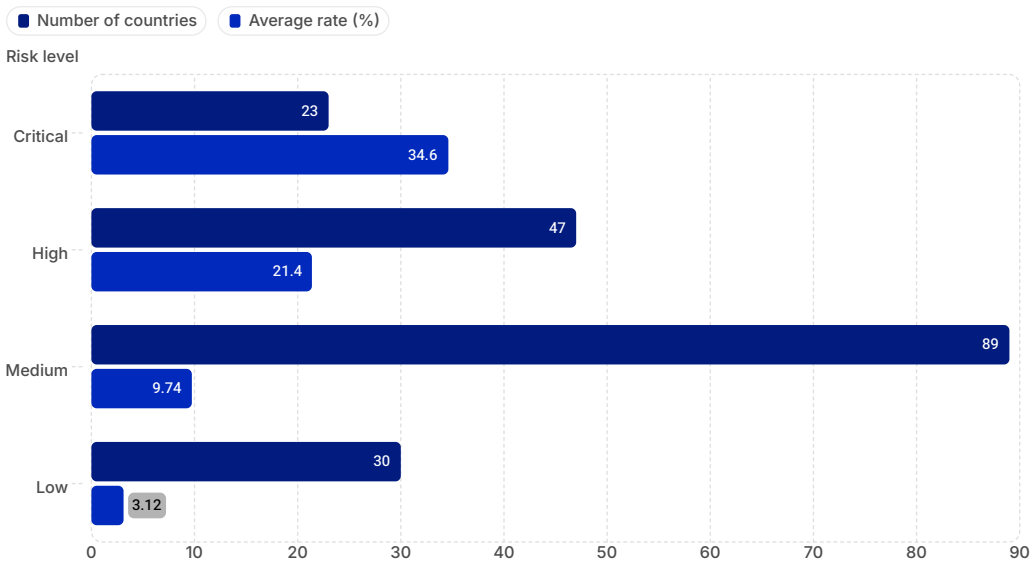
Key Result 1: Global Structural Level

Top 5 most fragile countries (2014–2024 average rate)

Country	Average rate
South Africa	42.18%
Eswatini	38.74%
Djibouti	36.92%
Namibia	35.61%
Gaza Strip	34.87%

Southern Africa and conflict zones concentrate the most extreme structural vulnerabilities — labor markets in a chronic state of crisis.

Global risk distribution in 2024



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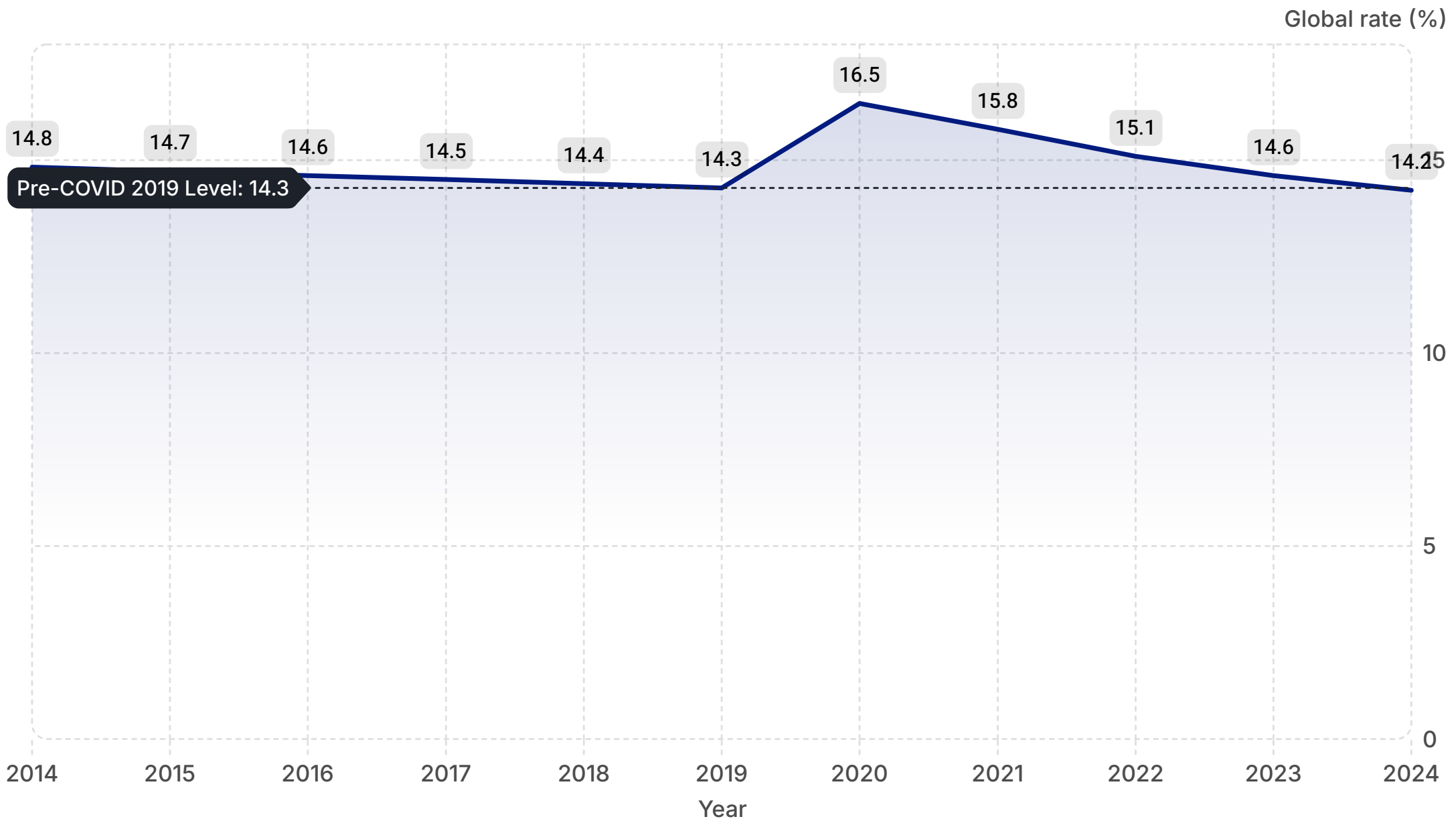
Key Result 2: Global Trend & COVID Shock



COVID Shock — Youth

+3.84 pts — The most affected segment. Young people experienced a shock 82% more intense than adults.

COVID Shock — Adults

+2.11 pts — Moderate but persistent impact on the adult labor market.

COVID Shock — Children

+1.93 pts — The least directly exposed segment, but vulnerable in the long term.

⚠️ In 2024, the global rate (14.22%) remains **above the pre-COVID 2019 level** (14.28%) — the recovery is incomplete.

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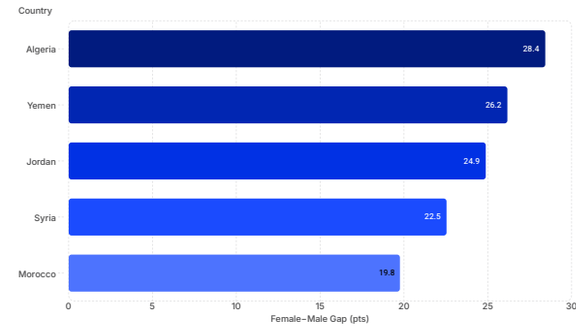
Key Result 3: Gender Inequality

Top 5 countries — Highest Gender Gap in 2024



Global view by segment in 2024

Youth — Women

22.84% vs Men: 18.63%

Gap: **+4.21 pts**

Adults — Women

10.47% vs Men: 7.92%

Gap: **+2.55 pts**

Gender inequality is **systemic across ALL segments** — women show a higher rate in every age group.

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Key Result 4: Youth Vulnerability

Countries where youth unemployment exceeds 40% in 2024 — more than one in two young people is without work.

1	**South Africa** Youth rate: **64.32%** Gap vs adults: **+36.18 pts**
2	**Eswatini** Youth rate: **61.47%** Gap vs adults: **+37.14 pts**
3	**Gaza Strip** Youth rate: **58.93%** Gap vs adults: **+36.52 pts**
4	**Namibia** Youth rate: **54.28%** Gap vs adults: **+34.41 pts**
5	**Libya** Youth rate: **49.83%** Gap vs adults: **+29.39 pts**

In these countries, youth is not simply suffering more — it is the **structural core of the problem**. These labor markets are generationally broken.

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Maximum Alert: 7 Priority Countries

Countries combining Youth Critical + worsening trend — These 7 countries combine the highest risk level AND a deterioration over 10 years.

🇦🇫 Afghanistan

+18.74 pts over 10 years

🇵🇸 Gaza Strip

+13.88 pts over 10 years

🇻🇪 Venezuela

+14.32 pts over 10 years

🇱🇧 Lebanon

+10.93 pts over 10 years

🇸🇩 Sudan

+9.76 pts over 10 years

🇭🇹 Haiti

+8.84 pts over 10 years

🇲🇿 Mozambique

+7.34 pts over 10 years

⚠️ Without structural intervention, their situation will continue to deteriorate **regardless of global conditions**.

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